

Package ‘nocturn’

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Title Sleep data filtering and visualisation

Version 1.0.1

Description This package helps importing, filtering and visualising sleep data.

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Suggests bslib,
devtools,
DT,
readr,
shinyjs,
shinyWidgets,
svglite,
testthat

Config/testthat/edition 3

Imports circular,
cli,
dplyr,
edfReader,
ggnewscale,
ggplot2,
lubridate,
rmarkdown,
readxl,
rlang,
scales,
shiny,
shinyalert,
stringr,
tibble,
tidyr

Depends R (>= 4.3)

LazyData true

Contents

check_suggests	3
chronotype	3
clean_epochs	4
clean_sessions	5
composite_phase_deviation	5
example_epochs	6
example_epochs_v1	7
example_sessions	8
example_sessions_v1	9
filter_by_age_range	10
filter_by_night_range	11
filter_by_sex	12
filter_epochs_from_sessions	13
get_epochs_summary	14
get_non_complying_sessions	15
get_removed_sessions	15
get_sessions_summary	16
group_epochs_by_night	17
group_sessions_by_night	18
interdaily_stability	19
load_batch	19
load_epochs	20
load_sessions	21
max_time	22
mean_time	22
min_time	23
nocturn	24
parse_date	24
parse_time	25
plot_actigram	25
plot_bedtimes_waketimes	26
plot_hypnogram	27
plot_sleep_bubbles	27
plot_sleep_clock	28
plot_sleep_spiral	29
plot_timeseries	29
plot_timeseries_sessions	30
read_edf_epochs	31
read_edf_sessions	32
remove_sessions_no_sleep	32
sd_time	33
select_devices	34
select_subjects	35
set_min_time_in_bed	36
set_session_sleep_onset_range	37
set_session_start_time_range	38

check_suggests

3

shift_times_by_12h

39

sleeptimes_boxplot

40

sleeptimes_density

40

sleeptimes_histogram

41

sleep_regularity_index

42

sleep_report

43

social_jet_lag

43

time_diff

44

time_to_hours

45

update_date

45

Index

47

check_suggests	Check whether suggested packages are installed
----------------	--

Description

Checks whether all the packages in Suggests are installed and prompts for install if not.

Usage

check_suggests()

Value

No return value, called for side effects

chronotype	Calculate the Chronotype
------------	--------------------------

Description

This function calculates the Chronotype metric based on the mid-sleep time If sleep duration on free days is greater than on workdays, it applies a correction as described in Roenneberg et al. (2019).

Usage

chronotype(sessions)

Arguments

sessions The sessions data frame

Details

This function uses columns:

- time_at_midsleep
- sleep_period
- is_workday

Value

The Chronotype value in hours

See Also

Other sleep metrics: [composite_phase_deviation\(\)](#), [interdaily_stability\(\)](#), [sleep_regularity_index\(\)](#), [social_jet_lag\(\)](#)

Examples

```
chronotype(example_sessions)
```

clean_epochs	<i>Clean epoch data</i>
--------------	-------------------------

Description

Clean epoch data

Usage

```
clean_epochs(epochs)
```

Arguments

epochs A dataframe containing the epoch data

Value

The cleaned-up epochs dataframe

Examples

```
epochs <- clean_epochs(nocturn::example_epochs)
```

clean_sessions	<i>Clean session data</i>
----------------	---------------------------

Description

Clean session data

Usage

```
clean_sessions(sessions)
```

Arguments

sessions A dataframe containing the session data

Value

The cleaned-up sessions dataframe

Examples

```
sessions <- clean_sessions(nocturn::example_sessions)
```

composite_phase_deviation	<i>Calculate Composite Phase Deviation (CPD)</i>
---------------------------	--

Description

This function calculates the Composite Phase Deviation (CPD) metric, used to measure the regularity of the sleep patterns.

Usage

```
composite_phase_deviation(sessions)
```

Arguments

sessions The sessions data frame

Details

This function uses columns:

- time_at_midsleep
- is_workday
- night

Value

The Composite Phase Deviation (CPD) value

See Also

Other sleep metrics: [chronotype\(\)](#), [interdaily_stability\(\)](#), [sleep_regularity_index\(\)](#), [social_jet_lag\(\)](#)

Examples

```
composite_phase_deviation(example_sessions)
```

example_epochs

Example Epoch data

Description

A data frame containing epoch data recorded by a Somnify device.

Usage

```
example_epochs
```

Format

example_epochs:

A data frame with 18,755 rows and 15 columns. Each row represents a time-point (or epoch) in a session. Epochs are 30 seconds long. The columns are as follows:

- `timestamp`: The time at which the epoch was recorded in UTC.
- `subject_id`: The ID of the subject.
- `signal_quality_mean`: The mean signal quality of the epoch.
- `movement_fast_mean`: The mean movement detected during the epoch.
- `movement_fast_nonzero_pct`
- `distance_mean`: the distance of the subject from the device in meters.
- `motion_data_count`: The number of data points in the epoch (30).
- `light_ambient_mean`: The ambient light level during the epoch.
- `sound_amplitude_mean`: The sound amplitude during the epoch.
- `temperature_ambient_mean`: The ambient temperature during the epoch.
- `humidity_mean`: The ambient humidity during the epoch.
- `pressure_mean`: The ambient pressure during the epoch.
- `indoor_air_quality_mean`: The indoor air quality during the epoch.
- `epoch_duration`: The precise duration of the epoch (seconds).
- `sleep_stage`: The sleep stage as established with the VT algorithm. They are encoded as numbers 0-5

Source

data-raw/example_epochs.csv

example_epochs_v1

Example Epoch data (Somnify API v1)

Description

A data frame containing epoch data recorded by a Somnify device.

Usage

example_epochs_v1

Format

example_epochs_v1:

A data frame with 1,373 rows and 16 columns. The corresponding session ID is contained in the file name. Each row represents a time-point (or epoch) in a session. Epochs are 30 seconds long. The columns are as follows:

- timestamp: The time at which the epoch was recorded in UTC.
- signal_quality_mean: The mean signal quality of the epoch.
- movement_fast_mean: The mean movement detected during the epoch.
- movement_fast_nonzero_pct
- distance_mean: the distance of the subject from the device in meters.
- motion_data_count: The number of data points in the epoch (30).
- light_ambient_mean: The ambient light level during the epoch.
- sound_amplitude_mean: The sound amplitude during the epoch.
- temperature_ambient_mean: The ambient temperature during the epoch.
- humidity_mean: The ambient humidity during the epoch.
- pressure_mean: The ambient pressure during the epoch.
- indoor_air_quality_mean: The indoor air quality during the epoch.
- epoch_duration: The precise duration of the epoch (seconds).
- sleep_stage: The sleep stage as established with the VT algorithm. They are encoded as numbers 0-5

Source

data-raw/SEtXSxcMEhYXKQAA.example_epochs_v1.csv

example_sessions	<i>Example Sessions data</i>
------------------	------------------------------

Description

A data frame containing sessions recorded by a Somnify device.

Usage

```
example_sessions
```

Format

`example_sessions`:

A data frame with 124 rows and 60 columns. Each row represents a session. Columns contain metadata about the session, including:

- `session_start`: The start time of the session in UTC.
- `session_end`: The end time of the session in UTC.
- `subject_id`: The ID of the subject.
- `device_serial_number`: The serial number of the device used.
- `time_at_sleep_onset`: The time at which the subject fell asleep.
- `time_at_wakeup`: The time at which the subject woke up. Columns also include various metrics averaged over the session, such as:
 - mean heart rate
 - mean respiration rate
- Finally, some columns contain environmental parameters, such as:
 - Temperature
 - Humidity
 - Light intensity
 - Noise level
 - Atmospheric pressure

Source

`data-raw/example_sessions.csv`

example_sessions_v1	<i>Example Sessions data (Somnify API v1)</i>
---------------------	---

Description

A data frame containing sessions recorded by a Somnify device.

Usage

example_sessions_v1

Format

example_sessions_v1:

A data frame with 87 rows and 70 columns. Each row represents a session. Columns contain metadata about the session, including:

- user_id: The ID of the recorded subject.
- sex: The sex of the recorded subject.
- birth_year: The year of birth of the recorded subject.
- session_start: The start time of the session in UTC.
- session_end: The end time of the session in UTC.
- time_at_sleep_onset: The time at which the subject fell asleep.
- time_at_wakeup: The time at which the subject woke up. Columns also include various metrics averaged over the session, such as:
 - mean heart rate
 - mean respiration rate
- Finally, some columns contain environmental parameters, such as:
 - Temperature
 - Humidity
 - Light intensity
 - Noise level
 - Atmospheric pressure

Source

data-raw/example_sessions_v1.csv

filter_by_age_range	<i>Filter sessions by age range</i>
---------------------	-------------------------------------

Description

Filter sessions by age range

Usage

```
filter_by_age_range(  
  sessions,  
  min_age = NULL,  
  max_age = NULL,  
  return_mask = FALSE  
)
```

Arguments

sessions	The sessions dataframe
min_age	The minimum age of the subjects (inclusive)
max_age	The maximum age of the subjects (inclusive)
return_mask	If TRUE, returns a logical vector indicating which sessions belong to subjects within the specified age range

Details

This function uses columns:

- birth_year

Value

The sessions dataframe with only the sessions that belong to subjects within the specified age range, or a logical vector if return_mask is TRUE

See Also

Other filtering: [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_sessions\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- filter_by_age_range(example_sessions_v1, min_age = 11, max_age = 18)
```

`filter_by_night_range` *Filter sessions for nights within a night range*

Description

Filter sessions for nights within a night range

Usage

```
filter_by_night_range(sessions, from_night, to_night, return_mask = FALSE)
```

Arguments

<code>sessions</code>	The sessions dataframe
<code>from_night</code>	The start night of the range (inclusive) in YYYY-MM-DD format
<code>to_night</code>	The end night of the range (inclusive) in YYYY-MM-DD format
<code>return_mask</code>	If TRUE, returns a logical vector indicating which sessions meet the night range requirement

Details

This function uses columns:

- `night`

Value

The sessions dataframe with only the sessions that fall within the specified night range, or a logical vector if `return_mask` is TRUE

See Also

Other filtering: [filter_by_age_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_sessions\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- filter_by_night_range(example_sessions, "2025-04-07", "2025-04-10")
```

filter_by_sex	<i>Filter by sex</i>
---------------	----------------------

Description

Filter by sex

Usage

```
filter_by_sex(sessions, sex, return_mask = FALSE)
```

Arguments

sessions	The sessions dataframe
sex	The sex to filter for (M, F, or NULL for both)
return_mask	If TRUE, returns a logical vector indicating which sessions belong to the specified sex

Details

This function uses columns:

- sex

Value

The sessions dataframe with only the sessions that belong to the specified sex, or a logical vector if return_mask is TRUE

See Also

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_epochs_from_sessions\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- filter_by_sex(example_sessions_v1, "M")
```

filter_epochs_from_sessions

Filter epochs based on session IDs

Description

Filter epochs based on session IDs

Usage

```
filter_epochs_from_sessions(epochs, sessions, return_mask = FALSE)
```

Arguments

epochs	The epochs dataframe
sessions	The sessions dataframe
return_mask	If TRUE, returns a logical vector indicating which epochs belong to the specified sessions

Details

This function uses sessions columns:

- session_start
- session_end And epoch columns:
- timestamp

Value

The epochs dataframe with only the epochs that belong to the specified sessions, or a logical vector if return_mask is TRUE

See Also

[filter_by_night_range\(\)](#) to filter sessions by night range.

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
# Apply filtering to sessions to keep specific nights, and filter epochs accordingly
filtered_sessions <- filter_by_night_range(example_sessions, "2025-04-07", "2025-04-10")
filtered_epochs <- filter_epochs_from_sessions(example_epochs, filtered_sessions)
```

get_epochs_summary	<i>Summarise epoch information</i>
--------------------	------------------------------------

Description

This function displays the number of sessions in the epoch data, as well as the start and end dates of the epoch data

Usage

```
get_epochs_summary(epochs)
```

Arguments

epochs	The epochs dataframe
--------	----------------------

Details

This function uses columns:

- timestamp
- session_id

Value

A single-row dataframe summarising epoch information

See Also

[get_sessions_summary\(\)](#) to summarise session information.

Other data tables: [get_non_complying_sessions\(\)](#), [get_removed_sessions\(\)](#), [get_sessions_summary\(\)](#)

Examples

```
get_epochs_summary(example_epochs)
```

`get_non_complying_sessions`*Get non-complying sessions (i.e. where there is more than one session on the same day)*

Description

Get non-complying sessions (i.e. where there is more than one session on the same day)

Usage

```
get_non_complying_sessions(sessions)
```

Arguments

`sessions` The sessions dataframe

Details

This function uses columns:

- `night`

Value

The sessions dataframe with only the sessions that are non-complying

See Also

Other data tables: [get_epochs_summary\(\)](#), [get_removed_sessions\(\)](#), [get_sessions_summary\(\)](#)

Examples

```
duplicate_sessions <- get_non_complying_sessions(example_sessions)
```

`get_removed_sessions` *Get a table of sessions that were removed during filtering*

Description

Get a table of sessions that were removed during filtering

Usage

```
get_removed_sessions(sessions, filtered_sessions)
```

Arguments

`sessions` The original sessions dataframe
`filtered_sessions` The filtered sessions dataframe

Details

This function uses columns:

- `id`
- `sleep_period`

Value

The sessions dataframe with only the sessions that were removed during filtering

See Also

Other data tables: [get_epochs_summary\(\)](#), [get_non_complying_sessions\(\)](#), [get_sessions_summary\(\)](#)

Examples

```
filtered_sessions <- set_session_start_time_range(example_sessions, "22:00", "06:00")
removed_sessions <- get_removed_sessions(example_sessions, filtered_sessions)
```

`get_sessions_summary` *Make a summary of session information*

Description

This function summarises session information, including the number of sessions, mean session length, mean time at sleep onset and wakeup, subject and device ID.

Usage

```
get_sessions_summary(sessions)
```

Arguments

`sessions` The sessions dataframe.

Details

This function uses columns:

- `time_at_sleep_onset`
- `time_at_wakeup`
- `time_in_bed`
- `sleep_period`

Value

A single-row dataframe summarizing session information.

See Also

[get_epochs_summary\(\)](#) to summarise epoch information.

Other data tables: [get_epochs_summary\(\)](#), [get_non_complying_sessions\(\)](#), [get_removed_sessions\(\)](#)

Examples

```
get_sessions_summary(example_sessions)
```

`group_epochs_by_night` *Create a grouping by night for epoch data*

Description

The function creates a new column `night` that groups the epochs by night. Timepoints before 12 PM are considered part of the previous night.

Usage

```
group_epochs_by_night(epochs)
```

Arguments

`epochs` The epochs dataframe

Details

This function uses columns:

- `timestamp`

Value

The epochs dataframe with the `night` column added

See Also

[group_sessions_by_night\(\)](#) to group session data by night.

Other time processing: [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
epochs <- group_epochs_by_night(example_epochs)
```

`group_sessions_by_night`*Create a grouping by night for session data*

Description

The function creates a new column `night` that groups the sessions by night depending on their start time. Sessions that start before 12 PM are considered part of the previous night.

Usage

```
group_sessions_by_night(sessions)
```

Arguments

<code>sessions</code>	The sessions dataframe
-----------------------	------------------------

Details

This function uses columns:

- `session_start`

Value

The sessions dataframe with the `night` column added

See Also

[group_epochs_by_night\(\)](#) to group epoch data by night.

Other time processing: [group_epochs_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
sessions <- group_sessions_by_night(example_sessions)
```

interdaily_stability	<i>Calculate Interdaily Stability (IS)</i>
----------------------	--

Description

This function calculates the Interdaily Stability (IS) metric from a binary awake/asleep variable

Usage

```
interdaily_stability(epochs)
```

Arguments

epochs	The epochs data frame
--------	-----------------------

Details

This function uses columns:

- timestamp
- is_asleep

Value

The Interdaily Stability (IS) value

See Also

Other sleep metrics: [chronotype\(\)](#), [composite_phase_deviation\(\)](#), [sleep_regularity_index\(\)](#), [social_jet_lag\(\)](#)

Examples

```
interdaily_stability(example_epochs)
```

load_batch	<i>Load session or epoch data in batch mode</i>
------------	---

Description

Load session or epoch data in batch mode

Usage

```
load_batch(
  folder_path = NULL,
  file_list = NULL,
  file_names = NULL,
  pattern = NULL,
  type = "sessions"
)
```

Arguments

folder_path	The path to the folder containing session files (do not use with file_list)
file_list	The list of file paths to load (do not use with folder_path)
file_names	An optional vector of file names corresponding to the files in file_list
pattern	An optional pattern to filter files in the folder
type	The type of data to load: "sessions" or "epochs"

Value

A dataframe containing the combined session data from all matching files in the folder

See Also

Other data loading: [load_epochs\(\)](#), [load_sessions\(\)](#), [read_edf_epochs\(\)](#), [read_edf_sessions\(\)](#)

load_epochs	<i>Load epoch data</i>
-------------	------------------------

Description

Load epoch data

Usage

```
load_epochs(epochs_file, file_name = NULL)
```

Arguments

epochs_file	The path to the epochs file
file_name	An optional file name to be recorded in the epochs table

Details

The function loads the epoch data from a file and groups the epochs by night. Supported formats: CSV, Excel, EDF.

Value

A dataframe containing the epoch data

See Also

Other data loading: [load_batch\(\)](#), [load_sessions\(\)](#), [read_edf_epochs\(\)](#), [read_edf_sessions\(\)](#)

load_sessions	<i>Load session data</i>
---------------	--------------------------

Description

Load session data

Usage

```
load_sessions(sessions_file)
```

Arguments

`sessions_file` The path to the sessions file

Details

The function loads the session data from a file Supported formats: CSV, Excel, EDF.

Value

A dataframe containing the session data

See Also

Other data loading: [load_batch\(\)](#), [load_epochs\(\)](#), [read_edf_epochs\(\)](#), [read_edf_sessions\(\)](#)

max_time	<i>Calculate the maximum time from 12pm to 12pm</i>
----------	---

Description

This function calculates the maximum time from a vector of time strings in the format "YYYY-MM-DD HH:MM:SS". It considers a time window from 12pm to 12pm the next day, so 11:00 is considered later than 13:00.

Usage

```
max_time(time_vector)
```

Arguments

time_vector A vector of time strings in the format "YYYY-MM-DD HH:MM:SS".

Value

A string representing the maximum time in the format "HH:MM".

See Also

[min_time\(\)](#) to calculate the minimum time in the same format.
Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
max_time(c("2025-04-08 23:00:00", "2025-04-09 01:00:00", "2025-04-09 02:30:00"))
```

mean_time	<i>Calculate the mean time from a vector of time strings</i>
-----------	--

Description

This function calculates the mean time from a vector of time strings in the format "YYYY-MM-DD HH:MM:SS".

Usage

```
mean_time(time_vector, unit = "HH:MM")
```

Arguments

time_vector	A vector of time strings in format "YYYY-MM-DD HH:MM:SS", "HH:MM:SS" or "HH:MM".
unit	The unit of time for the result. Can be "HH:MM" (default), "hour", "minute" or "second".

Value

A string representing the mean time in the format "HH:MM".

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
# Use on a vector of time strings representing full dates
time_vector <- c("2025-04-08 23:00:00", "2025-04-09 01:00:00")
mean_time(time_vector)

# Use on time-only strings
time_vector <- c("22:56", "01:32")
mean_time(time_vector)

# Use on a dataframe column
mean_time(example_sessions$time_at_sleep_onset)
```

min_time	<i>Calculate the minimum time from 12pm to 12pm</i>
----------	---

Description

This function calculates the minimum time from a vector of time strings in the format "YYYY-MM-DD HH:MM:SS". It considers a time window from 12pm to 12pm the next day, so 11:00 is considered later than 13:00.

Usage

```
min_time(time_vector)
```

Arguments

time_vector	A vector of time strings in the format "YYYY-MM-DD HH:MM:SS".
-------------	---

Value

A string representing the minimum time in the format "HH:MM".

See Also

`max_time()` to calculate the maximum time in the same format.
Other time processing: `group_epochs_by_night()`, `group_sessions_by_night()`, `max_time()`, `mean_time()`, `parse_date()`, `parse_time()`, `sd_time()`, `shift_times_by_12h()`, `time_diff()`, `time_to_hours()`, `update_date()`

Examples

```
min_time(c("2025-04-08 23:00:00", "2025-04-09 01:00:00", "2025-04-09 02:30:00"))
```

nocturn	<i>nocturn app</i>
---------	--------------------

Description

This function launches the nocturn app, a Shiny application for visualizing and analyzing sleep data.

Usage

```
nocturn()
```

parse_date	<i>Parse a vector of date strings into Date objects</i>
------------	---

Description

This function parses a vector of date strings into Date objects. All formats containing year, month, and day information are supported.

Usage

```
parse_date(date_vector)
```

Arguments

date_vector A vector of date strings

Value

A vector of Date objects

See Also

Other time processing: `group_epochs_by_night()`, `group_sessions_by_night()`, `max_time()`, `mean_time()`, `min_time()`, `parse_time()`, `sd_time()`, `shift_times_by_12h()`, `time_diff()`, `time_to_hours()`, `update_date()`

 parse_time

Parse a vector of time strings into POSIXct objects

Description

This function parses a vector of time strings into POSIXct objects. Supported formats include "YYYY-MM-DD HH:MM:SS", "YYYY-MM-DD HH:MM", "HH:MM:SS", and "HH:MM". Time-zone information is ignored.

Usage

```
parse_time(time_vector)
```

Arguments

time_vector A vector of time strings

Value

A vector of POSIXct objects

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

 plot_actigram

Plot an Actigram

Description

Generate an actigram from the Somnofy epoch data.

Usage

```
plot_actigram(epochs)
```

Arguments

epochs The epochs data frame

Details

This function uses columns:

- sleep_period
- signal_quality_mean
- sleep_stage

Value

A ggplot object representing the actigram

```
plot_bedtimes_waketimes
```

Plot bedtimes and waketimes

Description

Plot bedtimes and waketimes

Usage

```
plot_bedtimes_waketimes(sessions, groupby = "night", color_by = "default")
```

Arguments

sessions	The sessions dataframe
groupby	The grouping variable for the plot. Can be "night", "workday", or "weekday".
color_by	The variable to color the bars by. Can be "default" or any other column name in the sessions dataframe. Note that if color_by is anything else than "default", groupby will be set to "night".

Details

This function uses columns:

- night
- time_at_sleep_onset
- time_at_wakeup
- is_workday

Value

A ggplot graph showing the bedtimes and waketimes

plot_hypnogram	<i>Plot Hypnogram</i>
----------------	-----------------------

Description

Plot Hypnogram

Usage

```
plot_hypnogram(epochs)
```

Arguments

epochs	The epochs dataframe
--------	----------------------

Details

This function uses columns:

- timestamp
- sleep_stage

Value

A ggplot object showing the hypnogram as bars

See Also

Other plot epochs: [plot_sleep_spiral\(\)](#), [plot_timeseries\(\)](#)

plot_sleep_bubbles	<i>Plot Sleep Bubbles</i>
--------------------	---------------------------

Description

This function creates a bubble plot of sleep sessions, where the size and colour of the bubbles represents the sleep duration.

Usage

```
plot_sleep_bubbles(sessions, color_by = "default")
```

Arguments

sessions	The sessions dataframe.
color_by	The variable to color the bubbles by. Can be "default" or any other column name in the sessions dataframe.

Details

This function uses columns:

- sleep_period
- night

Value

A ggplot object containing the sleep bubbles graph.

See Also

Other plot sessions: [plot_sleep_clock\(\)](#), [plot_timeseries_sessions\(\)](#)

plot_sleep_clock	<i>Plot Sleep Clock</i>
------------------	-------------------------

Description

Plot Sleep Clock

Usage

```
plot_sleep_clock(sessions, color_by = "default")
```

Arguments

sessions	The sessions dataframe
color_by	The variable to color the segments by. Can be "default" or any other column name in the sessions dataframe.

Details

This function uses columns:

- time_at_sleep_onset
- time_at_wakeup
- night

Value

A ggplot object showing the sleep clock

See Also

Other plot sessions: [plot_sleep_bubbles\(\)](#), [plot_timeseries_sessions\(\)](#)

plot_sleep_spiral	<i>Plot Sleep Spiral</i>
-------------------	--------------------------

Description

Plot Sleep Spiral

Usage

```
plot_sleep_spiral(epochs, color_by = "default")
```

Arguments

epochs	The epochs dataframe
color_by	The variable to color the spiral by. Can be "default" or any other column name in the epochs dataframe.

Details

This function uses columns:

- timestamp
- is_asleep

Value

A ggplot object showing the sleep spiral

See Also

Other plot epochs: [plot_hypnogram\(\)](#), [plot_timeseries\(\)](#)

plot_timeseries	<i>Plot epoch time series data for a given variable</i>
-----------------	---

Description

Plot epoch time series data for a given variable

Usage

```
plot_timeseries(epochs, variable, color_by = "default", exclude_zero = FALSE)
```

Arguments

epochs	The epochs dataframe
variable	The variable to plot (e.g., "temperature_ambient_mean")
color_by	The variable to color the points by. Can be "default" or any other column name in the epochs dataframe.
exclude_zero	Logical, whether to exclude zero values from the plot (default: FALSE)

Details

This function uses columns:

- timestamp
- night

Value

A ggplot object

See Also

[plot_timeseries_sessions\(\)](#) to plot session data.

Other plot epochs: [plot_hypnogram\(\)](#), [plot_sleep_spiral\(\)](#)

plot_timeseries_sessions

Plot session time series data for a given variable

Description

Plot session time series data for a given variable

Usage

```
plot_timeseries_sessions(
  sessions,
  variable,
  color_by = "default",
  exclude_zero = FALSE
)
```

Arguments

sessions	The sessions dataframe
variable	The variable to plot (e.g., "time_at_sleep_onset")
color_by	The variable to color the points by. Can be "default" or any other column name in the sessions dataframe.
exclude_zero	Logical, whether to exclude zero values from the plot (default: FALSE)

Details

This function uses columns:

- night

Value

A ggplot object

See Also

[plot_timeseries\(\)](#) to plot epoch data.

Other plot sessions: [plot_sleep_bubbles\(\)](#), [plot_sleep_clock\(\)](#)

read_edf_epochs	<i>Read EDF Epochs</i>
-----------------	------------------------

Description

Read EDF Epochs

Usage

```
read_edf_epochs(file)
```

Arguments

file	The path to the EDF file
------	--------------------------

Details

The function reads the signals of the EDF file to extract epoch information. It must contain a column for timestamps and a column for sleep stage annotations.

Value

A dataframe containing the epoch data extracted from the EDF file signals

See Also

Other data loading: [load_batch\(\)](#), [load_epochs\(\)](#), [load_sessions\(\)](#), [read_edf_sessions\(\)](#)

read_edf_sessions	<i>Read EDF Sessions</i>
-------------------	--------------------------

Description

Read EDF Sessions

Usage

```
read_edf_sessions(file)
```

Arguments

file	The path to the EDF file
------	--------------------------

Details

The function reads the header of the EDF file to extract session information such as start time, duration, and calculates end time and midsleep time.

Value

A dataframe containing the session data extracted from the EDF file header

See Also

Other data loading: [load_batch\(\)](#), [load_epochs\(\)](#), [load_sessions\(\)](#), [read_edf_epochs\(\)](#)

remove_sessions_no_sleep	<i>Remove sessions with no sleep</i>
--------------------------	--------------------------------------

Description

Remove sessions with no sleep

Usage

```
remove_sessions_no_sleep(sessions, return_mask = FALSE)
```

Arguments

sessions	The sessions dataframe
return_mask	If TRUE, returns a logical vector indicating which sessions have a sleep period greater than 0

Details

This function uses columns:

- sleep_period

Value

The sessions dataframe with only the sessions that have a sleep period greater than 0, or a logical vector if return_mask is TRUE

See Also

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- remove_sessions_no_sleep(example_sessions)
```

sd_time

Calculate the circular standard deviation of a vector of times

Description

This function calculates the standard deviation of a vector of time strings, accounting for the circular nature of time (e.g., 23:59 is close to 00:00).

Usage

```
sd_time(time_vector, unit = "hour")
```

Arguments

time_vector	A vector of time strings in format "YYYY-MM-DD HH:MM:SS", "HH:MM:SS" or "HH:MM".
unit	The unit of time for the result. Can be "second", "minute", or "hour". Default is "hour".

Value

A numeric value representing the standard deviation in the specified unit.

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
sd_time(c("23:59", "00:01"))
```

select_devices	<i>Select devices by ID</i>
----------------	-----------------------------

Description

Select devices by ID

Usage

```
select_devices(sessions, device_ids, return_mask = FALSE)
```

Arguments

sessions	The sessions dataframe
device_ids	The device IDs to select
return_mask	If TRUE, returns a logical vector indicating which sessions were recorded by the specified devices

Details

This function uses columns:

- device_id

Value

The sessions dataframe with only the sessions recorded by the specified devices, or a logical vector if return_mask is TRUE

See Also

[select_subjects\(\)](#) to select sessions by subject ID.
Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session_start\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- select_devices(example_sessions, c("VTGVSRTTHCA"))
```

select_subjects	Select subjects by ID
-----------------	-----------------------

Description

Select subjects by ID

Usage

```
select_subjects(sessions, subject_ids, return_mask = FALSE)
```

Arguments

- | | |
|-------------|---|
| sessions | The sessions dataframe |
| subject_ids | The subject IDs to select |
| return_mask | If TRUE, returns a logical vector indicating which sessions belong to the specified subjects <ul style="list-style-type: none">• subject_id |

Details

This function uses columns:

Value

The sessions dataframe with only the sessions that belong to the specified subjects, or a logical vector if return_mask is TRUE

See Also

[select_devices\(\)](#) to select sessions by device ID.

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- select_subjects(example_sessions, c("sub_01JNDH3Z5NP0PSV82NFBGPV31X"))
```

set_min_time_in_bed	<i>Set minimum time in bed</i>
---------------------	--------------------------------

Description

Set minimum time in bed

Usage

```
set_min_time_in_bed(sessions, min_time_in_bed, return_mask = FALSE)
```

Arguments

sessions	The sessions dataframe
min_time_in_bed	The minimum time in bed in hours
return_mask	If TRUE, return a logical vector indicating which sessions meet the minimum time in bed requirement

Details

This function uses columns:

- time_in_bed

Value

The sessions dataframe with only the sessions that meet the minimum time in bed requirement, or a logical vector if return_mask is TRUE

See Also

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_session_sleep_onset_range\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- set_min_time_in_bed(example_sessions, 2)
```

```
set_session_sleep_onset_range
    Set sleep onset time range
```

Description

Set sleep onset time range

Usage

```
set_session_sleep_onset_range(
  sessions,
  from_time,
  to_time,
  return_mask = FALSE
)
```

Arguments

sessions	The sessions dataframe
from_time	Include sessions where sleep started after this time (in format HH:MM)
to_time	Include sessions where sleep started before this time (in format HH:MM)
return_mask	If TRUE, returns a logical vector indicating which sessions meet the sleep onset time range requirement

Details

This function uses columns:

- time_at_sleep_onset

Value

The sessions dataframe with only the sessions where sleep started within the specified time range, or a logical vector if return_mask is TRUE

See Also

[set_session_start_time_range\(\)](#) to filter sessions based on start time.

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_start_time_range\(\)](#)

Examples

```
filtered_sessions <- set_session_sleep_onset_range(example_sessions, "22:00", "06:00")
```

```
set_session_start_time_range
```

Set session start time range

Description

Set session start time range

Usage

```
set_session_start_time_range(sessions, from_time, to_time, return_mask = FALSE)
```

Arguments

<code>sessions</code>	The sessions dataframe
<code>from_time</code>	Include sessions that started after this time (in format HH:MM)
<code>to_time</code>	Include sessions that started before this time (in format HH:MM)
<code>return_mask</code>	If TRUE, returns a logical vector indicating which sessions meet the time range requirement

Details

This function uses columns:

- `session_start`

Value

The sessions dataframe with only the sessions that started within the specified time range, or a logical vector if `return_mask` is TRUE

See Also

[set_session_sleep_onset_range\(\)](#) to filter sessions based on sleep onset time.

Other filtering: [filter_by_age_range\(\)](#), [filter_by_night_range\(\)](#), [filter_by_sex\(\)](#), [filter_epochs_from_session\(\)](#), [remove_sessions_no_sleep\(\)](#), [select_devices\(\)](#), [select_subjects\(\)](#), [set_min_time_in_bed\(\)](#), [set_session_sleep_onset_range\(\)](#)

Examples

```
filtered_sessions <- set_session_start_time_range(example_sessions, "22:00", "06:00")
```

shift_times_by_12h	<i>Shift times to break at 12 pm</i>
--------------------	--------------------------------------

Description

This function shifts times so that the day starts at 12 PM. This is useful for plotting night data

Usage

```
shift_times_by_12h(times)
```

Arguments

times	A vector of times in POSIXct format, character convertible to POSIXct, or numerical (in hours).
-------	---

Value

A vector of times in POSIXct format (or numerical if numerical provided as input) shifted to start at 12 PM

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
# Shift a vector of times in HH:MM format
shift_times_by_12h(c("02:30", "16:00"))
#> "14:30" "04:00"

# Shift times in YYYY-MM-DD HH:MM:SS format
shift_times_by_12h(c("2025-04-08 23:00:00", "2025-04-09 01:00:00"))
#> "2025-04-08 11:00" "2025-04-09 13:00"

# Shift sessions start times to start at 12 PM
shifted_times <- shift_times_by_12h(example_sessions$session_start)

# Use dplyr::mutate to directly add the shifted times to a dataframe
epochs <- example_epochs |>
  dplyr::mutate(shifted_time = shift_times_by_12h(timestamp))
```

sleeptimes_boxplot	<i>Plot boxplots for sleep onset, midsleep, and wakeup times</i>
--------------------	--

Description

Plot boxplots for sleep onset, midsleep, and wakeup times

Usage

```
sleeptimes_boxplot(sessions, circular = FALSE)
```

Arguments

sessions	The sessions dataframe
circular	Whether to output a circular plot (default FALSE)

Details

This function uses columns:

- time_at_sleep_onset
- time_at_wakeup
- time_at_midsleep

Value

A ggplot object with three horizontal boxplots (onset, midsleep, wakeup)

sleeptimes_density	<i>Plot density curves for sleep onset, midsleep, and wakeup times with a dashed line showing the median</i>
--------------------	--

Description

Plot density curves for sleep onset, midsleep, and wakeup times with a dashed line showing the median

Usage

```
sleeptimes_density(sessions, adjust = 1, circular = FALSE)
```

Arguments

sessions	The sessions dataframe
adjust	The bandwidth adjustment for the density estimate (default 1)
circular	Whether to output a circular plot (default FALSE)

Details

This function uses columns:

- time_at_sleep_onset
- time_at_wakeup
- time_at_midsleep

Value

A ggplot object with three overlaid density curves (sleep onset, midsleep, wakeup)

sleeptimes_histogram *Plot histograms for sleep onset, midsleep, and wakeup times*

Description

Plot histograms for sleep onset, midsleep, and wakeup times

Usage

```
sleeptimes_histogram(sessions, binwidth = 0.25, circular = FALSE)
```

Arguments

sessions	The sessions dataframe
binwidth	The width of the bins for the histogram (default 0.25)
circular	Whether to output a circular plot (default FALSE)

Details

This function uses columns:

- time_at_sleep_onset
- time_at_wakeup
- time_at_midsleep

Value

A ggplot object with three overlaid histograms (sleep onset, midsleep, wakeup)

`sleep_regularity_index`*Calculate the Sleep Regularity Index (SRI)*

Description

The Sleep Regularity Index (SRI) is a measure of the regularity of sleep patterns. It is calculated as the percentage of epochs where the sleep state remains the same after 24 hours.

Usage

```
sleep_regularity_index(epochs)
```

Arguments

epochs	The epochs data frame
--------	-----------------------

Details

This function uses columns:

- timestamp
- is_asleep

Value

The Sleep Regularity Index (SRI) value

See Also

Other sleep metrics: [chronotype\(\)](#), [composite_phase_deviation\(\)](#), [interdaily_stability\(\)](#), [social_jet_lag\(\)](#)

Examples

```
sleep_regularity_index(example_epochs)
```

sleep_report	<i>Generate a patient sleep report in PDF format</i>
--------------	--

Description

This function generates a sleep report in PDF format using an R Markdown template. It is designed to work with Somnofy data, so some values may not be available when using other data sources such as GGIR.

Usage

```
sleep_report(sessions, title = "", output_file = "Sleep_report.pdf")
```

Arguments

sessions	The sessions dataframe
title	The title of the report. Default is an empty string.
output_file	Path for the output PDF. Default is "Sleep_report.pdf"

Details

This function uses columns:

- night
- time_at_sleep_onset
- time_at_wakeup
- time_at_midsleep
- sleep_onset_latency

social_jet_lag	<i>Calculate Social Jet Lag</i>
----------------	---------------------------------

Description

This function calculates the Social Jet Lag (SJM) metric as the difference in mid-sleep times between workdays and free days.

Usage

```
social_jet_lag(sessions)
```

Arguments

sessions	The sessions data frame
----------	-------------------------

Details

This function uses columns:

- time_at_midsleep
- is_workday

Value

The Social Jet Lag (SJL) value in hours

See Also

Other sleep metrics: [chronotype\(\)](#), [composite_phase_deviation\(\)](#), [interdaily_stability\(\)](#), [sleep_regularity_index\(\)](#)

Examples

```
social_jet_lag(example_sessions)
```

time_diff	<i>Compute the forward time difference from t1 to t2 (wrapping at 24)</i>
-----------	---

Description

This function returns the time from t1 to t2, always moving forward on the clock. For example, from 07:00 to 22:00 is 15 hours, from 22:00 to 07:00 is 9 hours.

Usage

```
time_diff(t1, t2, unit = "hour")
```

Arguments

- | | |
|------|--|
| t1 | First time (character, POSIXct, or numeric hour) |
| t2 | Second time (character, POSIXct, or numeric hour) |
| unit | The unit of time. Can be "second", "minute", or "hour". Default is "hour". |

Value

The forward difference in the specified unit (numeric, always positive, $0 \leq x < 24$)

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_to_hours\(\)](#), [update_date\(\)](#)

Examples

```
time_diff("07:00", "22:00") # 15
time_diff("22:00", "07:00") # 9
time_diff("07:00", "22:00", unit = "minute") # 540
```

time_to_hours	<i>Convert time vector to numeric hours</i>
---------------	---

Description

This function converts a vector of time strings or POSIXct objects to numeric hours.

Usage

```
time_to_hours(time_vector)
```

Arguments

time_vector A vector of time strings

Details

See [parse_time\(\)](#) for supported time formats.

Value

A numeric vector representing the time in hours

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [update_date\(\)](#)

update_date	<i>Update the date component of a POSIXct time object</i>
-------------	---

Description

This function updates the date component of a POSIXct time object while preserving the time component.

Usage

```
update_date(time, date)
```

Arguments

time	A POSIXct time object or a character string convertible to POSIXct
date	A Date object or a character string convertible to Date

Value

A POSIXct time object with the updated date

See Also

Other time processing: [group_epochs_by_night\(\)](#), [group_sessions_by_night\(\)](#), [max_time\(\)](#), [mean_time\(\)](#), [min_time\(\)](#), [parse_date\(\)](#), [parse_time\(\)](#), [sd_time\(\)](#), [shift_times_by_12h\(\)](#), [time_diff\(\)](#), [time_to_hours\(\)](#)

Index

- * **data loading**
 - load_batch, [19](#)
 - load_epochs, [20](#)
 - load_sessions, [21](#)
 - read_edf_epochs, [31](#)
 - read_edf_sessions, [32](#)
- * **data tables**
 - get_epochs_summary, [14](#)
 - get_non_complying_sessions, [15](#)
 - get_removed_sessions, [15](#)
 - get_sessions_summary, [16](#)
- * **datasets**
 - example_epochs, [6](#)
 - example_epochs_v1, [7](#)
 - example_sessions, [8](#)
 - example_sessions_v1, [9](#)
- * **filtering**
 - filter_by_age_range, [10](#)
 - filter_by_night_range, [11](#)
 - filter_by_sex, [12](#)
 - filter_epochs_from_sessions, [13](#)
 - remove_sessions_no_sleep, [32](#)
 - select_devices, [34](#)
 - select_subjects, [35](#)
 - set_min_time_in_bed, [36](#)
 - set_session_sleep_onset_range, [37](#)
 - set_session_start_time_range, [38](#)
- * **plot epochs**
 - plot_hypnogram, [27](#)
 - plot_sleep_spiral, [29](#)
 - plot_timeseries, [29](#)
- * **plot sessions**
 - plot_sleep_bubbles, [27](#)
 - plot_sleep_clock, [28](#)
 - plot_timeseries_sessions, [30](#)
- * **sleep metrics**
 - chronotype, [3](#)
 - composite_phase_deviation, [5](#)
 - interdaily_stability, [19](#)
 - sleep_regularity_index, [42](#)
 - social_jet_lag, [43](#)
- * **time processing**
 - group_epochs_by_night, [17](#)
 - group_sessions_by_night, [18](#)
 - max_time, [22](#)
 - mean_time, [22](#)
 - min_time, [23](#)
 - parse_date, [24](#)
 - parse_time, [25](#)
 - sd_time, [33](#)
 - shift_times_by_12h, [39](#)
 - time_diff, [44](#)
 - time_to_hours, [45](#)
 - update_date, [45](#)
- check_suggests, [3](#)
- chronotype, [3](#), [6](#), [19](#), [42](#), [44](#)
- clean_epochs, [4](#)
- clean_sessions, [5](#)
- composite_phase_deviation, [4](#), [5](#), [19](#), [42](#), [44](#)
- example_epochs, [6](#)
- example_epochs_v1, [7](#)
- example_sessions, [8](#)
- example_sessions_v1, [9](#)
- filter_by_age_range, [10](#), [11–13](#), [33–38](#)
- filter_by_night_range, [10](#), [11](#), [12](#), [13](#), [33–38](#)
- filter_by_night_range(), [13](#)
- filter_by_sex, [10](#), [11](#), [12](#), [13](#), [33–38](#)
- filter_epochs_from_sessions, [10–12](#), [13](#), [33–38](#)
- get_epochs_summary, [14](#), [15–17](#)
- get_epochs_summary(), [17](#)
- get_non_complying_sessions, [14](#), [15](#), [16](#), [17](#)

get_removed_sessions, [14](#), [15](#), [15](#), [17](#)
 get_sessions_summary, [14–16](#), [16](#)
 get_sessions_summary(), [14](#)
 group_epochs_by_night, [17](#), [18](#), [22–25](#), [33](#),
 [39](#), [44–46](#)
 group_epochs_by_night(), [18](#)
 group_sessions_by_night, [17](#), [18](#), [22–25](#),
 [33](#), [39](#), [44–46](#)
 group_sessions_by_night(), [17](#)
 interdaily_stability, [4](#), [6](#), [19](#), [42](#), [44](#)

 load_batch, [19](#), [21](#), [31](#), [32](#)
 load_epochs, [20](#), [20](#), [21](#), [31](#), [32](#)
 load_sessions, [20](#), [21](#), [21](#), [31](#), [32](#)

 max_time, [17](#), [18](#), [22](#), [23–25](#), [33](#), [39](#), [44–46](#)
 max_time(), [24](#)
 mean_time, [17](#), [18](#), [22](#), [22](#), [24](#), [25](#), [33](#), [39](#),
 [44–46](#)
 min_time, [17](#), [18](#), [22](#), [23](#), [23](#), [24](#), [25](#), [33](#), [39](#),
 [44–46](#)
 min_time(), [22](#)

 nocturn, [24](#)

 parse_date, [17](#), [18](#), [22–24](#), [24](#), [25](#), [33](#), [39](#),
 [44–46](#)
 parse_time, [17](#), [18](#), [22–24](#), [25](#), [33](#), [39](#), [44–46](#)
 parse_time(), [45](#)
 plot_actigram, [25](#)
 plot_bedtimes_waketimes, [26](#)
 plot_hypnogram, [27](#), [29](#), [30](#)
 plot_sleep_bubbles, [27](#), [28](#), [31](#)
 plot_sleep_clock, [28](#), [28](#), [31](#)
 plot_sleep_spiral, [27](#), [29](#), [30](#)
 plot_timeseries, [27](#), [29](#), [29](#)
 plot_timeseries(), [31](#)
 plot_timeseries_sessions, [28](#), [30](#)
 plot_timeseries_sessions(), [30](#)

 read_edf_epochs, [20](#), [21](#), [31](#), [32](#)
 read_edf_sessions, [20](#), [21](#), [31](#), [32](#)
 remove_sessions_no_sleep, [10–13](#), [32](#),
 [34–38](#)

 sd_time, [17](#), [18](#), [22–25](#), [33](#), [39](#), [44–46](#)
 select_devices, [10–13](#), [33](#), [34](#), [35–38](#)
 select_devices(), [35](#)
 select_subjects, [10–13](#), [33](#), [34](#), [35](#), [36–38](#)

 select_subjects(), [34](#)
 set_min_time_in_bed, [10–13](#), [33–35](#), [36](#), [37](#),
 [38](#)
 set_session_sleep_onset_range, [10–13](#),
 [33–36](#), [37](#), [38](#)
 set_session_sleep_onset_range(), [38](#)
 set_session_start_time_range, [10–13](#),
 [33–37](#), [38](#)
 set_session_start_time_range(), [37](#)
 shift_times_by_12h, [17](#), [18](#), [22–25](#), [33](#), [39](#),
 [44–46](#)
 sleep_regularity_index, [4](#), [6](#), [19](#), [42](#), [44](#)
 sleep_report, [43](#)
 sleeptimes_boxplot, [40](#)
 sleeptimes_density, [40](#)
 sleeptimes_histogram, [41](#)
 social_jet_lag, [4](#), [6](#), [19](#), [42](#), [43](#)

 time_diff, [17](#), [18](#), [22–25](#), [33](#), [39](#), [44](#), [45](#), [46](#)
 time_to_hours, [17](#), [18](#), [22–25](#), [33](#), [39](#), [44](#), [45](#),
 [46](#)

 update_date, [17](#), [18](#), [22–25](#), [33](#), [39](#), [44](#), [45](#), [45](#)